University of Science and Technology in Zewail City

Department of Computer and Information Engineering

City Smart Hospital Network Design

CIE 447 / CIE 367 — Computer Networks Project

Submitted by:

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1. Team Contribution Statement

1.1 Contribution Details

- **Ahmed Essam — VLANs + DHCP:** Responsible for configuring all VLANs on every switch, setting up trunk and access ports, and configuring DHCP pools on Admin, Medical, and OutPatient routers. Verified that all end devices received correct IP addresses via DHCP.
- **Omar Tamer— OSPF + Routing:** Responsible for configuring Single-Area OSPF (Area 0) on all internal routers, verifying neighbor adjacencies, and ensuring all subnets were advertised and reachable across zones.
- **Mohamed Gadwal — NAT + ISP Connectivity:** Responsible for configuring NAT Overload (PAT) on CoreData router, setting up the default static route to the ISP, configuring the ISP router, and verifying external internet access from internal hosts.
- **Omar Ali — Topology Design + Cabling:** Responsible for the physical design and implementation of the network topology in Cisco Packet Tracer, including all router-to-router connections, switch-to-router trunk links, and serial DCE cable configurations.
- **Amr Emad — Report + Testing:** Responsible for conducting all verification tests (ping, traceroute, web browsing, simulation mode captures), collecting screenshots, and writing the project report.

2. Addressing Scheme & Justification

2.1 Complete IP Addressing Table

The hospital was assigned the private Class B block `172.16.0.0/16`. Fixed Length Subnet Masking (FLSM) was applied, dividing the block into `/24` subnets. The third octet matches the VLAN ID for clarity.

Table 2.1: Complete IP Addressing Scheme

Zone	**VLAN**	**Department**	**Network**	**Gateway**	**Broadcast**
Admin	10	Patient Records	172.16.10.0/24	172.16.10.1	172.16.10.255
Admin	20	Finance	172.16.20.0/24	172.16.20.1	172.16.20.255
Admin	30	HR	172.16.30.0/24	172.16.30.1	172.16.30.255
Medical	40	General Ward	172.16.40.0/24	172.16.40.1	172.16.40.255
Medical	50	ICU	172.16.50.0/24	172.16.50.1	172.16.50.255
Outpatient	60	Consultation	172.16.60.0/24	172.16.60.1	172.16.60.255
Outpatient	70	Guest Wi-Fi	172.16.70.0/24	172.16.70.1	172.16.70.255
Data Center	80	Servers	172.16.80.0/24	172.16.80.1	172.16.80.255
Radiology	90	Remote Clinic	172.16.90.0/24	172.16.90.1	172.16.90.255

Table 2.2: Point-to-Point Router Links

Link	**Network**	**Router 1 IP**	**Router 2 IP**
Admin ↔ CoreData	172.16.100.0/24	172.16.100.1	172.16.100.2
Admin ↔ Medical (Serial)	172.16.110.0/24	172.16.110.1	172.16.110.2
Medical ↔ CoreData	172.16.120.0/24	172.16.120.1	172.16.120.2
OutPatient ↔ CoreData	172.16.130.0/24	172.16.130.1	172.16.130.2
Admin ↔ Radiology (Serial)	172.16.140.0/24	172.16.140.1	172.16.140.2
CoreData ↔ ISP (Serial)	203.0.113.0/30	203.0.113.1	203.0.113.2

Table 2.3: Static IP Assignments for Servers

Device	IP Address	Subnet Mask	Gateway	DNS
Web Server (Intranet)	172.16.80.10	255.255.255.0	172.16.80.1	172.16.80.20
DNS Server	172.16.80.20	255.255.255.0	172.16.80.1	172.16.80.20
ISP Web Server	8.8.8.10	255.255.255.0	8.8.8.1	—

2.2 Subnetting Justification

2.2.1 Why is /24 Suitable for All Departments?

A /24 subnet provides 254 usable host addresses per subnet. This is sufficient for all departments in the hospital:

- The largest department is the General Ward with 120 devices — well within 254 addresses.
- Using /24 subnets simplifies configuration and troubleshooting.
- The third octet matches the VLAN ID, making the addressing scheme intuitive and easy to manage.

2.2.2 Unused IP Addresses (Wastage)

Table 2.4: IP Address Wastage Analysis

Department	Devices	Usable IPs	Wasted IPs
Patient Records	60	254	194
Finance	30	254	224
HR	15	254	239
General Ward	120	254	134
ICU	40	254	214
Consultation	50	254	204
Guest Wi-Fi	30	254	224
Servers	10	254	244
Radiology	20	254	234
Total	**375**	**2286**	**1911**

Approximately **1911 IP addresses** are unused across all department subnets, representing significant wastage of about 83.6%.

2.2.3 FLSM vs VLSM Comparison

Table 2.5: FLSM vs VLSM Comparison

Aspect	**FLSM (Used)**	**VLSM (Alternative)**
Address Efficiency	Low — same size for all subnets regardless of need	High — subnet sized to exact department need
Configuration Complexity	Simple and straightforward	More complex planning required
Scalability	Easy to add new subnets	Requires careful planning to avoid overlap
Troubleshooting	Easy — consistent pattern	Harder — varying subnet sizes
IP Wastage	High (1911 addresses wasted)	Minimal wastage
Suitability	Good for uniform or small networks	Better for large enterprise networks

3. Network Foundations

3.1 LAN and WAN Boundaries

3.1.1 LAN Segments

The Local Area Network (LAN) encompasses all internal hospital zones:

- **Zone 1 — Administration LAN:** Switch0 connects HR, Patient Records, and Finance PCs. Admin router handles inter-VLAN routing via Router-on-a-Stick on `GigabitEthernet0/1`.
- **Zone 2 — Medical LAN:** SwitchMedical connects Ward and ICU PCs. Medical router handles sub-interfaces on `GigabitEthernet0/1`.
- **Zone 3 — Outpatient LAN:** Switch2 connects Consultation PC and Access Point for wireless guests. OutPatient router handles sub-interfaces on `GigabitEthernet0/1`.
- **Zone 4 — Data Center LAN:** Switch3 connects the Web Server and DNS Server to CoreData router via VLAN 80.

3.1.2 WAN Segment

The Wide Area Network (WAN) boundary exists at the Serial link between **CoreData_R** (`203.0.113.1`) and **ISP_R** (`203.0.113.2`) using the public IP block `203.0.113.0/30`. This link uses a Serial DCE cable and represents the demarcation point between the private hospital network and the public internet.

3.2 How VLANs Separate Broadcast Domains

VLANs operate at Layer 2 (Data Link Layer) and logically segment the network into separate broadcast domains. Without VLANs, all devices on a switch share the same broadcast domain, meaning a broadcast from one device reaches all others — causing unnecessary traffic and security risks.

By assigning each department to a distinct VLAN:

- A broadcast from a device in VLAN 10 (Patient Records) is confined to VLAN 10 and never reaches VLAN 20 (Finance) or VLAN 30 (HR).

- Each switch port is configured as either an **access port** (assigned to one VLAN) or a **trunk port** (carries multiple VLANs using 802.1Q tagging).
- The trunk port connecting each switch to its router carries all VLAN traffic tagged, allowing the router's sub-interfaces to route between VLANs (Router-on-a-Stick method).
- Inter-VLAN communication requires routing through the router, providing an opportunity for access control and policy enforcement.

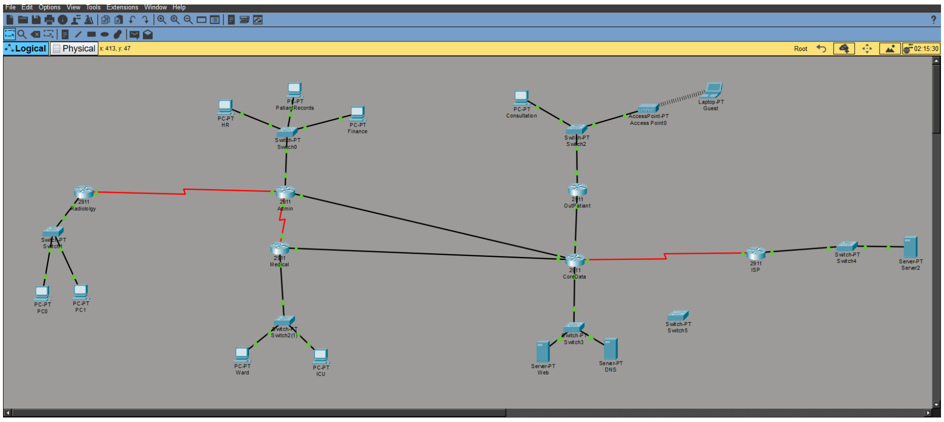

Figure 3.1: Successful ping before VLAN segmentation

4. Medium Comparison

4.1 Wired vs Wireless in Zone 3 (Outpatient Clinic)

Zone 3 contains two distinct network segments that use different transmission media:

Table 4.1: Wired vs Wireless Comparison in Zone 3

Aspect	**Consultation Rooms (Wired)**	**Guest Wi-Fi (Wireless)**
VLAN	VLAN 60	VLAN 70
Medium	Copper UTP (Straight-Through)	IEEE 802.11 (Wi-Fi)
Device	PC connected via FastEthernet	Laptop connected via Access Point
Switch Port	Access port (Fa1/1) on Switch2	Access port (Fa2/1) on Switch2
Speed	Up to 100 Mbps (FastEthernet)	Up to 54 Mbps (802.11g in PT-AP)
Security	Physical access required	SSID broadcast; less physically secure
Interference	None (shielded cable)	Subject to RF interference
Mobility	Fixed — cable required	Mobile — roam within AP range
Configuration	Switchport access VLAN 60	SSID set to GuestWiFi; VLAN 70
DHCP	Served by OutPatient_R pool VLAN60	Served by OutPatient_R pool VLAN70
Broadcast Domain	Separate from VLAN 70	Separate from VLAN 60

4.1.1 Key Configuration Differences

Wired (Consultation — VLAN 60):

```
Switch2(config)# interface FastEthernet1/1
Switch2(config-if)# switchport mode access
Switch2(config-if)# switchport access vlan 60
```

Wireless (Guest Wi-Fi — VLAN 70):

```
! Switch port to Access Point
Switch2(config)# interface FastEthernet2/1
Switch2(config-if)# switchport mode access
Switch2(config-if)# switchport access vlan 70

! Access Point configuration
! SSID: GuestWiFi
! Laptop connects wirelessly to Access Point
```

Both VLANs are served by the OutPatient router's sub-interfaces and receive DHCP leases from separate pools, ensuring traffic isolation between consultation staff and guest users.

5. Verification, Failover & Design Reasoning

5.1 Connectivity Verification

5.1.1 Ping Tests — Internal Servers

The following screenshots demonstrate successful ping connectivity from edge PCs to internal servers.

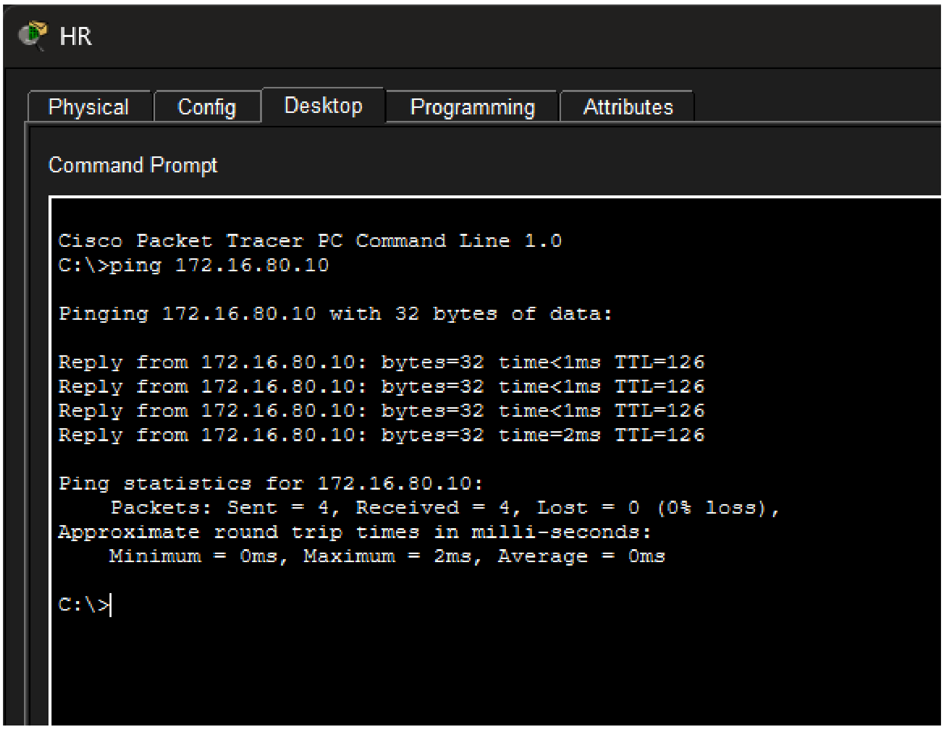

Figure 5.1: Successful ping before VLAN segmentation

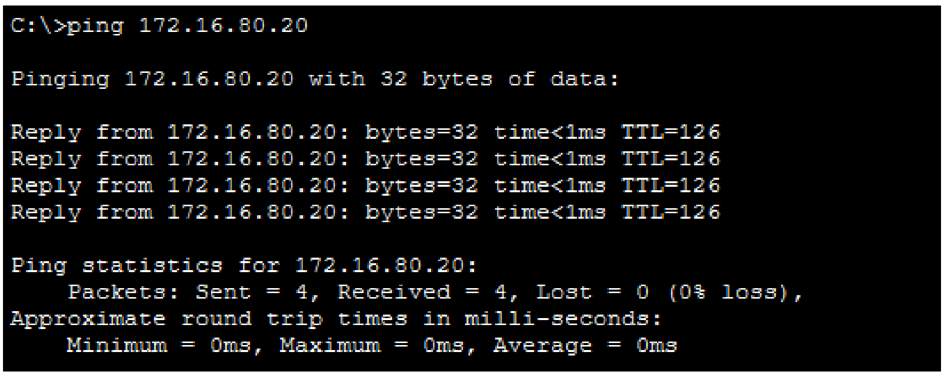

Figure 5.2: Ping from HR PC (VLAN 30) to DNS Server (172.16.80.20)

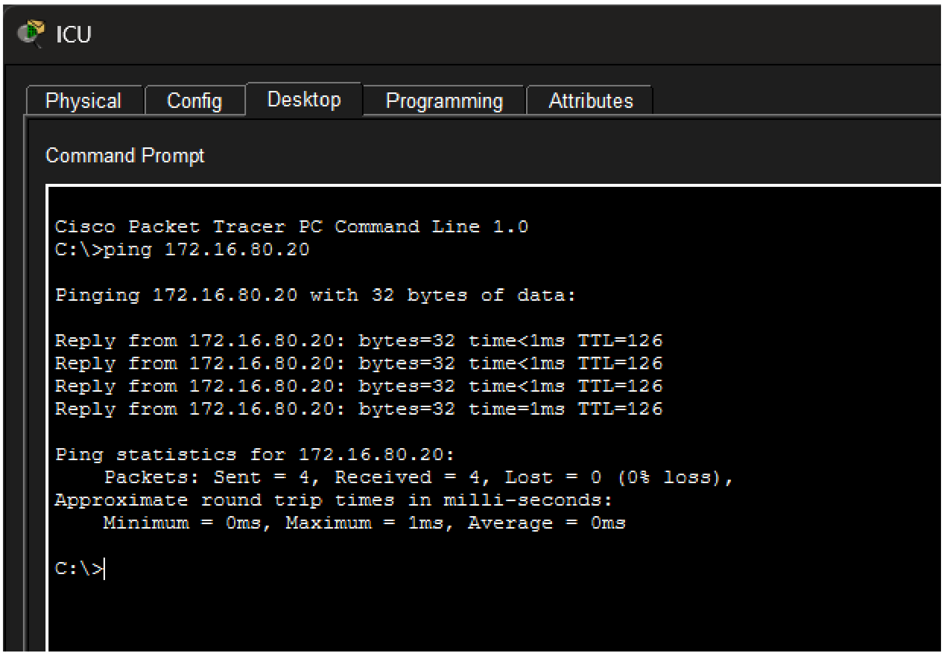

Figure 5.3: Ping from ICU PC to Web Server

5.1.2 Ping Tests — External Server

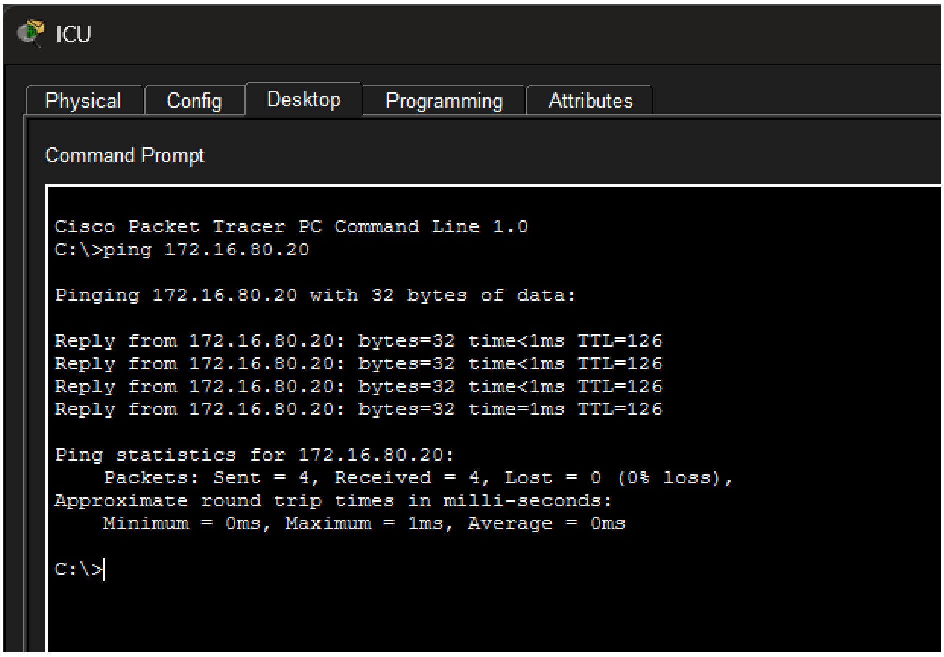

Figure 5.4: Ping from Internal PC to External ISP Web Server (8.8.8.10) via NAT

5.1.3 Web Browsing Verification

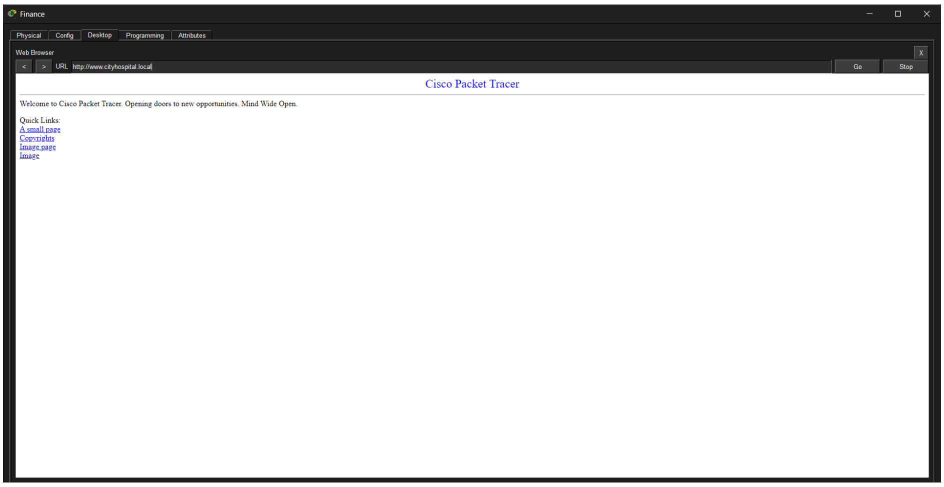

Figure 5.5: Successful Browsing of www.cityhospital.local from Internal PC

Figure 5.6: Successful Browsing of www.google.com (External) via NAT

5.2 Failover Testing (Phase 2 — Part B)

5.2.1 Traceroute Before Cable Cut

Figure 5.7: Traceroute from ICU PC to Data Center — Primary Path (Before Failure)

ICU PC → Medical_R → CoreData_R → Web Server

5.2.2 Traceroute After Cable Cut

Figure 5.8: Traceroute from ICU PC to Data Center — Backup Path (After Failure)

The backup path after the failure was:

$$\text{ICU PC} \rightarrow \text{Medical\_R} \rightarrow \text{Admin\_R} \rightarrow \text{CoreData\_R} \rightarrow \text{Web Server}$$

5.3 Design Reasoning

5.3.1 Why was the Backup Link placed between Medical_R and Admin_R?

The backup Serial DCE link was implemented between **Medical_R** and **Admin_R** (rather than a second direct link from Medical_R to CoreData_R) for the following reasons:

1. **Redundancy through path diversity:** Connecting Medical_R directly to CoreData_R again would only protect against a single cable failure but would not provide a truly diverse path. Routing through Admin_R creates a completely different physical path through the network.

2. **Resource efficiency:** Adding a second direct link to CoreData_R would consume an additional GigabitEthernet port on CoreData, which is already fully utilized. The Serial link to Admin_R uses the Serial module instead.

3. **OSPF cost-based routing:** OSPF assigns a higher cost to Serial links compared to GigabitEthernet links. This ensures the Serial backup path is only used when the primary GigabitEthernet path fails — OSPF automatically prefers the faster primary link during normal operation.

4. **Network resilience design principle:** The backup link ensures that even if the primary Medical_R ↔ CoreData_R fiber is severed, the Medical zone remains reachable through Admin_R, maintaining hospital operations.

5.3.2 Why Did Traffic Change Path After the Failure?

When the primary GigabitEthernet link between Medical_R and CoreData_R was deleted:

1. **OSPF Dead Timer:** OSPF detected the neighbor loss after the dead timer expired (40 seconds, or immediately with Fast Forward Time in Packet Tracer).
2. **LSA Flooding:** Medical_R flooded a new Link State Advertisement (LSA) indicating the loss of the direct link to CoreData_R.
3. **SPF Recalculation:** All routers ran the Shortest Path First (SPF) algorithm using Dijkstra's algorithm to recalculate the best paths.
4. **New Best Path:** The only remaining path from Medical_R to CoreData_R was via Admin_R (through the Serial backup link), so OSPF installed this as the new route.
5. **Traceroute Confirmation:** The traceroute output after the failure shows an additional hop through Admin_R (172.16.110.1 / 172.16.100.1), confirming OSPF reconvergence to the backup path.

6. Protocol Analysis

6.1 Traffic Capture in Simulation Mode

Using Cisco Packet Tracer's Simulation Mode, a web browser request to `www.cityhospital.local` was initiated from an edge PC. The resulting packet capture revealed two distinct protocol exchanges: a **DNS query** and an **HTTP request**.

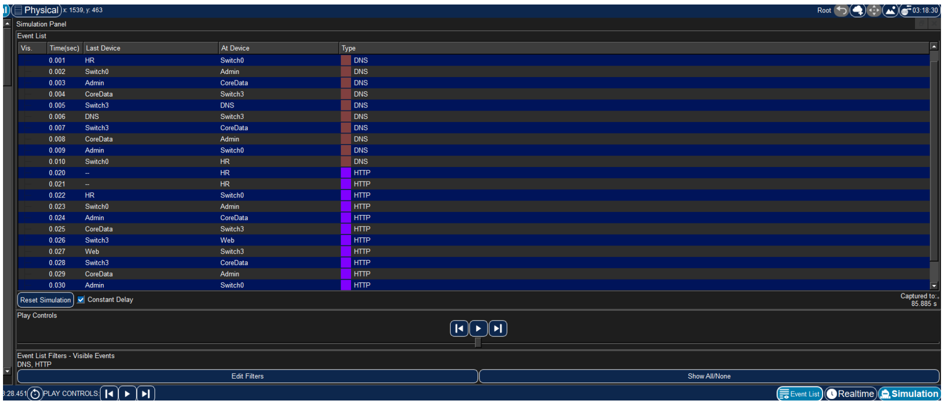

Figure 6.1: Packet Capture in Simulation Mode — DNS and HTTP Packets Visible

6.2 DNS Request Analysis

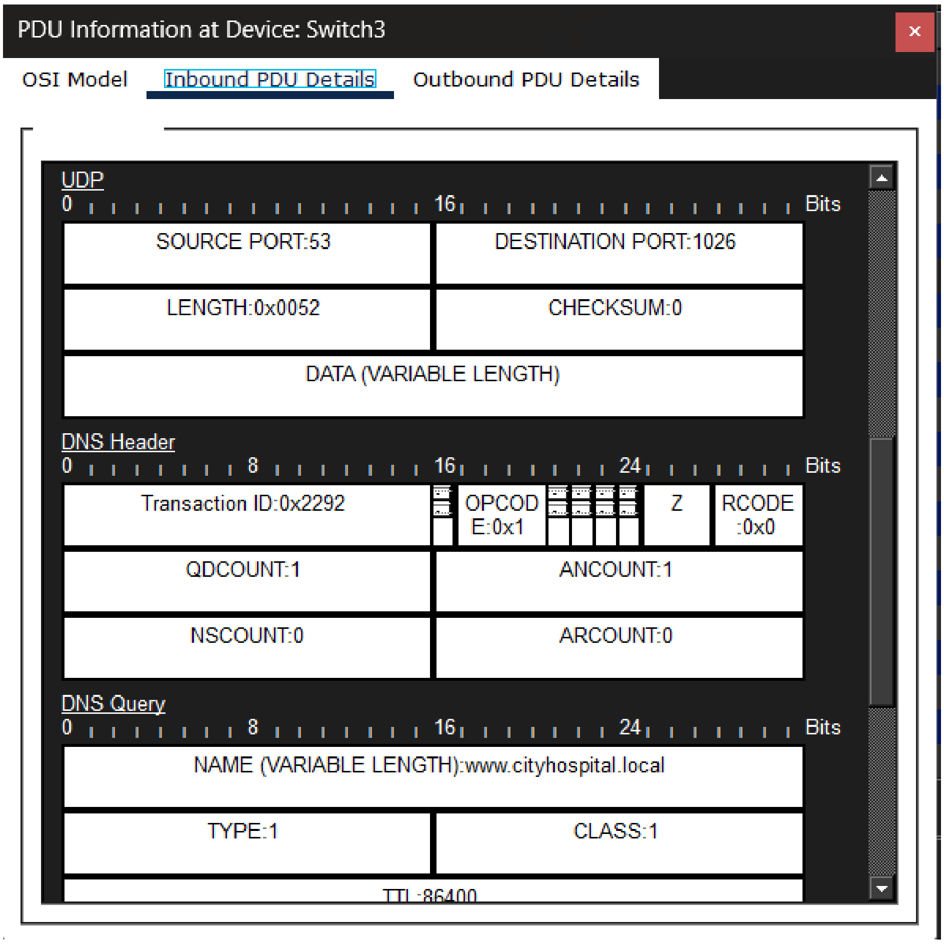

Figure 6.2: PDU Details of DNS packet — showing UDP port 53

6.2.1 DNS Uses UDP — Why?

DNS uses **UDP (User Datagram Protocol)** on **port 53** for the following reasons:

- **Speed:** DNS queries are small, simple request-response transactions. UDP has no connection setup overhead, making it significantly faster than TCP for this purpose.
- **Low overhead:** UDP does not perform handshaking, sequencing, or acknowledgment — ideal for short messages where the cost of establishing a TCP connection would outweigh the benefit.
- **Stateless:** DNS queries are independent transactions. If a DNS response is lost, the client simply retransmits the query — no TCP state management is needed.
- **Sufficient reliability:** For small DNS packets (typically under 512 bytes), UDP is reliable enough. The DNS application layer handles retransmission if needed.

6.3 HTTP Request Analysis

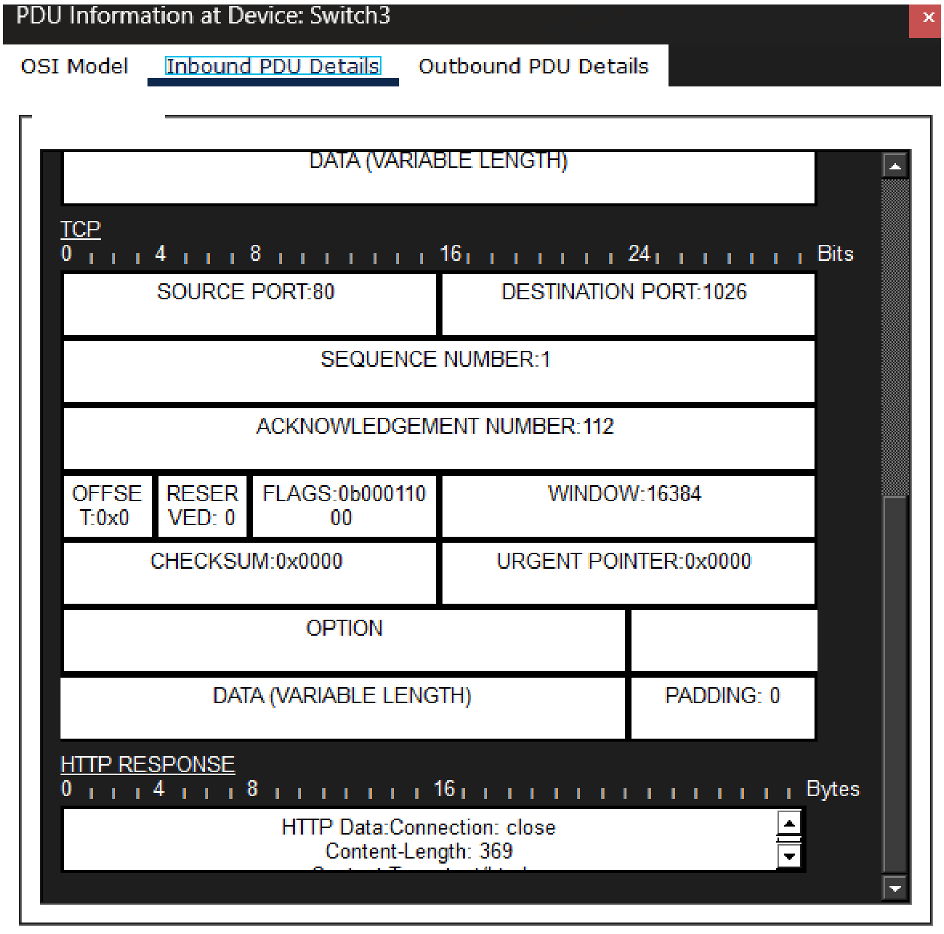

Figure 6.3: HTTP Request PDU Details — Transport Layer: TCP Port 80

6.3.1 HTTP Uses TCP — Why?

HTTP uses **TCP (Transmission Control Protocol)** on **port 80** for the following reasons:

- **Reliability:** Web pages consist of multiple large objects (HTML, images, scripts). TCP guarantees delivery of every byte in the correct order through acknowledgments and retransmission.
- **Connection-oriented:** TCP establishes a connection via the three-way handshake (SYN, SYN-ACK, ACK) before data transfer, ensuring both parties are ready.
- **Flow control:** TCP prevents the sender from overwhelming the receiver using a sliding window mechanism.
- **Error detection:** TCP checksums detect corruption, and lost segments are

retransmitted automatically.

- **Data integrity:** For web content, delivering corrupted or incomplete data would result in broken pages — TCP's reliability guarantees a complete, correct response.

6.4 DNS vs HTTP Comparison Summary

Table 6.1: DNS vs HTTP Protocol Comparison

Property	**DNS**	**HTTP**
Transport Protocol	UDP	TCP
Port Number	53	80
Connection Type	Connectionless	Connection-oriented
Reliability	Application-level retry	TCP guaranteed delivery
Handshake	None	3-way handshake (SYN/SYN-ACK/ACK)
Typical Payload Size	Small (under 512 bytes)	Large (HTML, images, scripts)
Latency	Very low	Higher (due to handshake)
Use Case	Name resolution only	Full web page transfer
Why appropriate	Fast lookup, no state needed	Reliable delivery of complex content

A. Router Configuration Scripts

A.1 Admin Router

```
hostname Admin
!
interface GigabitEthernet0/1
 no shutdown
!
interface GigabitEthernet0/1.10
 encapsulation dot1Q 10
 ip address 172.16.10.1 255.255.255.0
!
interface GigabitEthernet0/1.20
 encapsulation dot1Q 20
 ip address 172.16.20.1 255.255.255.0
!
interface GigabitEthernet0/1.30
 encapsulation dot1Q 30
 ip address 172.16.30.1 255.255.255.0
!
interface GigabitEthernet0/0
 ip address 172.16.100.1 255.255.255.0
 no shutdown
!
interface Serial0/3/0
 ip address 172.16.110.1 255.255.255.0
 clock rate 64000
 no shutdown
!
ip dhcp excluded-address 172.16.10.1
ip dhcp pool VLAN10
 network 172.16.10.0 255.255.255.0
 default-router 172.16.10.1
 dns-server 172.16.80.20
!
ip dhcp excluded-address 172.16.20.1
ip dhcp pool VLAN20
 network 172.16.20.0 255.255.255.0
 default-router 172.16.20.1
 dns-server 172.16.80.20
!
ip dhcp excluded-address 172.16.30.1
```

```
ip dhcp pool VLAN30
 network 172.16.30.0 255.255.255.0
 default-router 172.16.30.1
 dns-server 172.16.80.20
!
router ospf 1
 network 172.16.10.0 0.0.0.255 area 0
 network 172.16.20.0 0.0.0.255 area 0
 network 172.16.30.0 0.0.0.255 area 0
 network 172.16.100.0 0.0.0.255 area 0
 network 172.16.110.0 0.0.0.255 area 0
```

A.2 Medical Router

```
hostname Medical
!
interface GigabitEthernet0/1
 no shutdown
!
interface GigabitEthernet0/1.40
 encapsulation dot1Q 40
 ip address 172.16.40.1 255.255.255.0
!
interface GigabitEthernet0/1.50
 encapsulation dot1Q 50
 ip address 172.16.50.1 255.255.255.0
!
interface GigabitEthernet0/0
 ip address 172.16.120.1 255.255.255.0
 no shutdown
!
interface Serial0/3/0
 ip address 172.16.110.2 255.255.255.0
 no shutdown
!
ip dhcp excluded-address 172.16.40.1
ip dhcp pool VLAN40
 network 172.16.40.0 255.255.255.0
 default-router 172.16.40.1
 dns-server 172.16.80.20
!
ip dhcp excluded-address 172.16.50.1
ip dhcp pool VLAN50
 network 172.16.50.0 255.255.255.0
 default-router 172.16.50.1
 dns-server 172.16.80.20
```

```
!
router ospf 1
 network 172.16.40.0 0.0.0.255 area 0
 network 172.16.50.0 0.0.0.255 area 0
 network 172.16.110.0 0.0.0.255 area 0
 network 172.16.120.0 0.0.0.255 area 0
```

A.3 OutPatient Router

```
hostname OutPatient
!
interface GigabitEthernet0/1
 no shutdown
!
interface GigabitEthernet0/1.60
 encapsulation dot1Q 60
 ip address 172.16.60.1 255.255.255.0
!
interface GigabitEthernet0/1.70
 encapsulation dot1Q 70
 ip address 172.16.70.1 255.255.255.0
!
interface GigabitEthernet0/0
 ip address 172.16.130.1 255.255.255.0
 no shutdown
!
ip dhcp excluded-address 172.16.60.1
ip dhcp pool VLAN60
 network 172.16.60.0 255.255.255.0
 default-router 172.16.60.1
 dns-server 172.16.80.20
!
ip dhcp excluded-address 172.16.70.1
ip dhcp pool VLAN70
 network 172.16.70.0 255.255.255.0
 default-router 172.16.70.1
 dns-server 172.16.80.20
!
router ospf 1
 network 172.16.60.0 0.0.0.255 area 0
 network 172.16.70.0 0.0.0.255 area 0
 network 172.16.130.0 0.0.0.255 area 0
```

A.4 CoreData Router

```
hostname CoreData
!
interface GigabitEthernet0/0
 ip address 172.16.100.2 255.255.255.0
 ip nat inside
 no shutdown
!
interface GigabitEthernet0/1
 ip address 172.16.130.2 255.255.255.0
 ip nat inside
 no shutdown
!
interface GigabitEthernet0/2
 ip address 172.16.120.2 255.255.255.0
 ip nat inside
 no shutdown
!
interface FastEthernet0/2/0
 switchport access vlan 80
 no shutdown
!
interface Vlan80
 ip address 172.16.80.1 255.255.255.0
 ip nat inside
 no shutdown
!
interface Serial0/3/0
 ip address 203.0.113.1 255.255.255.252
 ip nat outside
 no shutdown
!
ip route 0.0.0.0 0.0.0.0 203.0.113.2
access-list 1 permit 172.16.0.0 0.0.255.255
ip nat inside source list 1 interface Serial0/3/0 overload
!
router ospf 1
 network 172.16.80.0 0.0.0.255 area 0
 network 172.16.100.0 0.0.0.255 area 0
 network 172.16.120.0 0.0.0.255 area 0
 network 172.16.130.0 0.0.0.255 area 0
 default-information originate
```

A.5 ISP Router

```
hostname ISP
```

```
!
interface Serial0/3/0
 ip address 203.0.113.2 255.255.255.252
 no shutdown
!
interface GigabitEthernet0/0
 ip address 8.8.8.1 255.255.255.0
 no shutdown
!
ip route 172.16.0.0 255.255.0.0 203.0.113.1
```

A.6 Radiology Router (Phase 2 — Part A)

```
hostname Radiology
!
interface GigabitEthernet0/0
 no shutdown
!
interface GigabitEthernet0/0.90
 encapsulation dot1Q 90
 ip address 172.16.90.1 255.255.255.0
!
interface Serial0/3/0
 ip address 172.16.140.2 255.255.255.0
 no shutdown
!
ip dhcp excluded-address 172.16.90.1
ip dhcp pool VLAN90
 network 172.16.90.0 255.255.255.0
 default-router 172.16.90.1
 dns-server 172.16.80.20
!
router ospf 1
 network 172.16.90.0 0.0.0.255 area 0
 network 172.16.140.0 0.0.0.255 area 0
```